

PSimPy - Predictive and probabilistic simulation

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Chair of Methods for Model-based Development in Computational Engineering

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Want to build unified, robust and sustainable
UQ workflows? ^[1]

PSimPy



Highlights

Robust GP emulator for your simulation

Save precious compute time!

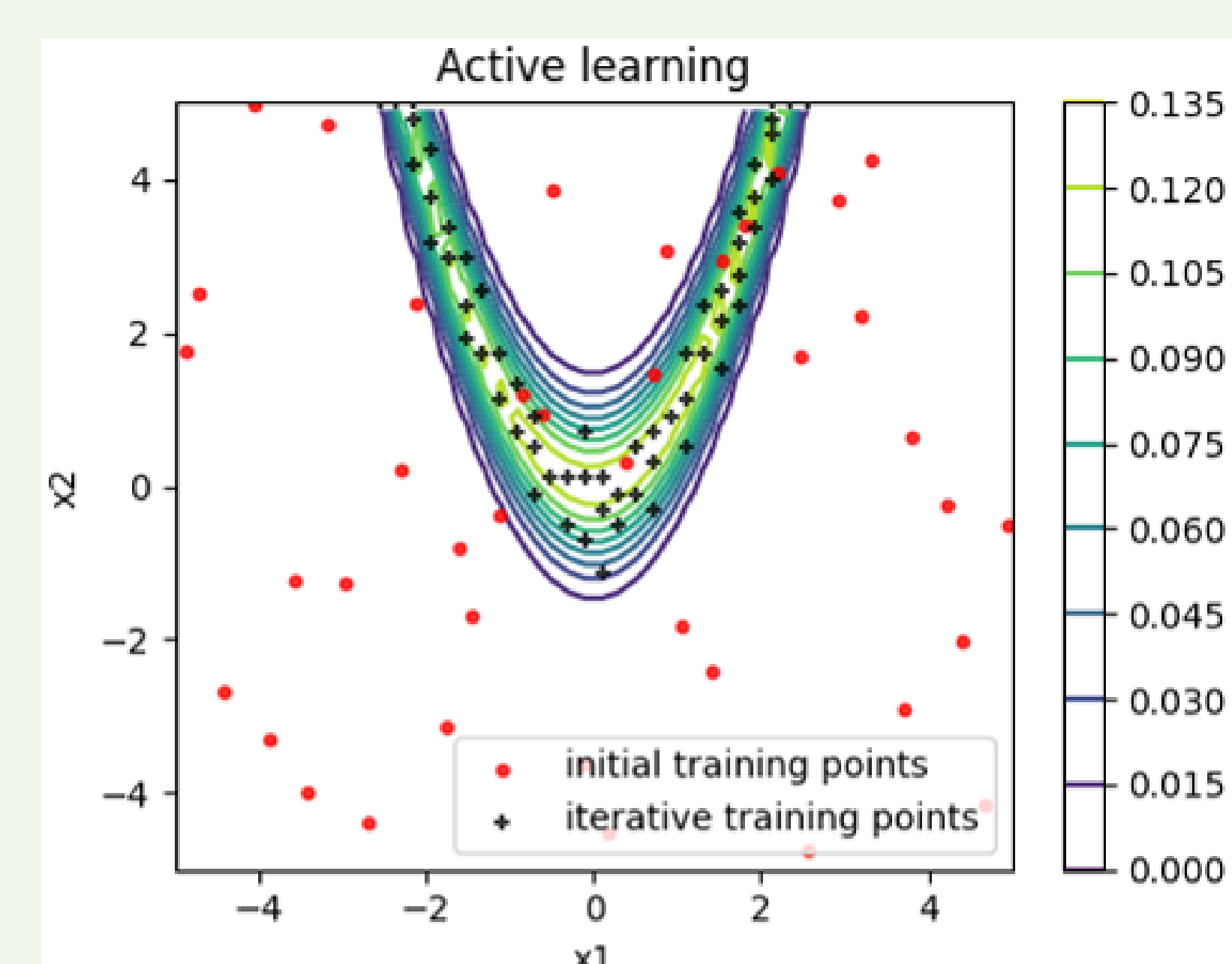
	training	evaluation	2500 evals
simulator	-	32 min	56 days
emulator	4.5 days	0.05 sec	2 min

- Gaussian Process Emulation
- Bayesian Calibration
- Global Sensitivity Analysis
- Efficient Sampling
- Active Learning

In Action

Quantify uncertainty. Identify key drivers.

Calibrate with data.



Open Source and Pythonic

[1] H. Zhao, A. Yildiz, N. Bagherinejad, and J. Kowalski, "PSimPy: GP emulation-based sensitivity analysis, uncertainty quantification and calibration of landslide simulators," in *14th International Conference on Applications of Statistics and Probability in Civil Engineering, ICASP14*, Dublin, Ireland, 2023.